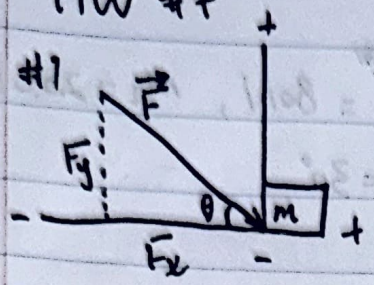
HW #7

#1



Given: $F = 80.0N$, $F_g = mg$

$m = 20.0kg$, F_i = impeding force

$\theta = 30°$, $F_x = \vec{F}\cos\theta$

a)

$$\cancel{\vec{F}_{netx} = \vec{F}_x + \vec{F}_i}$$
$$\cancel{0 = \vec{F}_x + \vec{F}_i}$$
$$\cancel{\vec{F}_i = -\vec{F}_x}$$
$$\cancel{= -F\cos 30°}$$
$$\cancel{= -80N \cdot \cos 30°}$$
$$\cancel{= -69.28N}$$

b)

$$\cancel{\vec{F}_{netx} = \vec{F}_x + \vec{F}_i}$$
$$\cancel{m\vec{a}_x = \vec{F}_x + \vec{F}_i}$$
$$\cancel{\vec{F}_x = m\vec{a}_x - \vec{F}_i}$$
$$\cancel{F}$$

a)

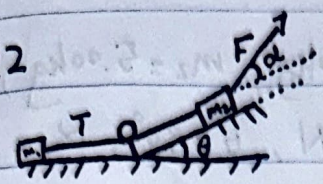
$$F_{netx} = F_x + F_i$$
$$0 = F_x + F_i$$
$$F_i = -F_x$$
$$= -\vec{F}\cos\theta$$
$$= -80.0N \cdot \cos 30°$$
$$= -69.28N$$

b)

$$F_x = \vec{F}\cos\theta$$
$$\vec{F} = \frac{F_x}{\cos\theta}$$

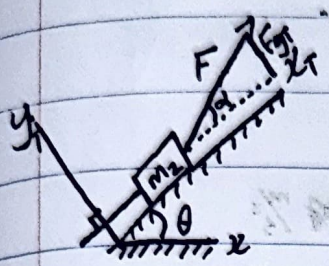
$$F_{netx} = F_x + F_i$$
$$ma_x = F_x + F_i$$
$$F_x = ma_x - F_i$$
$$\vec{F} = \frac{ma_x - F_i}{\cos\theta}$$
$$= \frac{20.0kg \cdot 1.25 m/s^2 + 69.28N}{\cos 30°}$$
$$= \cancel{\frac{44.28N}{\cos 30°}} + \frac{94.28N}{\cos 30°}$$
$$= 108.86N$$

#2



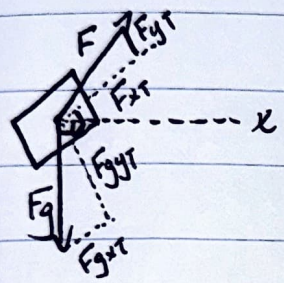
Given: $m_1 = 2.00\,kg$, $m_2 = 400\,kg$, $m_a = m_1 + m_2$

$\theta = 25°$, $\alpha = 15°$, $F = 26.0\,N$



∵ there is no movement along y_T axis,

$\therefore \overline{F_{netyT}} = \overline{F_{yT}} + \overline{F_{gyT}} = 0$



$\overline{F_{netxT}} = F\cos\alpha = 26.0N \cdot \cos 15° = 25.11N$

$\overline{F_{netxT}} = \overline{F_{xT}} + \overline{F_{gxT}}$

$\overline{F_{gxT}} = F_g \cdot \sin\alpha = mg \cdot \sin 15° = -10.16N$

$\overline{F_{netxT}} = 25.11N - 10.16N$

$= 14.95N$

$\overline{F_{netxT}} = m_a a_{xT}$

$14.95N = 6.00kg\ a_{xT}$

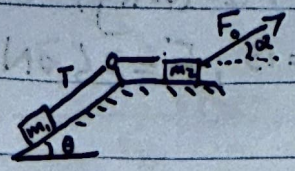
$a_{xT} = 2.49\ m/s^2$

$T = m_1 a_x$

$= 2kg \cdot 2.49\ m/s^2$

$= 4.98N$

#3



Given: $m_1 = 3.00\,kg$, $m_2 = 5.00\,kg$, $F_o = 10.0\,N$, $\theta = 30°$, $\alpha = 20°$

$m_a = m_1 + m_2$

$\vec{F_o} = m_a\vec{a}$

$\vec{a} = \frac{\vec{F_o}}{m_a}$

$= \frac{10.0\,N}{8.00\,kg}$

$= 1.25\,m/s^2$

$a_x = \vec{a}\cos\alpha$

$= 1.25\,m/s^2 \cdot \cos 20°$

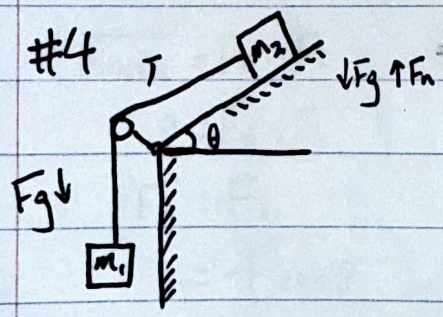
$= 1.17\,m/s^2$

$T = m_1 a_x$

$= 3.00\,kg \cdot 1.17\,m/s^2$

$= 3.52\,N$

#4



Given: $m_1 = 1.15\,kg$, $m_2 = 2.00\,kg$, $\theta = 20°$, $m_a = m_1 + m_2$

$\because M_{2g}F_{gm_1} > M_{2g}F_{gym_2}$

$\therefore \vec{F_{net}}$ is downward

$\vec{F_{netg}} = F_{gm_1} - F_{gym_2}$

$= -4.570\,N$

$F = m_a a$

$a = \frac{F}{m_a}$

$= -\frac{4.570\,N}{3.15\,kg}$

$= -1.451\,m/s^2$

$T = |m_1 a|$

$= 1.15\,kg \cdot -1.451\,m/s^2$

$= 1.668\,N$

$F_{gm_1} = m_1 g$

$= 1.15\,kg \cdot -9.81\,m/s^2$

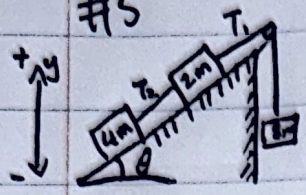
$= -11.28\,N$

$F_{gym_2} = m_2 \cdot g \cdot \sin\theta$

$= 2.00\,kg \cdot -9.81\,m/s^2 \cdot \sin 20°$

$= -6.710\,N$

#5



$g = -9.81\ m/s^2$

~~2) $M_s = 4M + 2M$, $M_d = 8M$, $M_2 = 4M+2M+8M$~~

~~$F_{gmsy} = g \cdot \sin\theta \cdot (4M+2M)$~~

~~$F_{gmdy} = g \cdot 1 \cdot 8M$~~

~~$\because 1 \geq \sin\theta$ & $8M > (4M+2M)$~~

~~$\therefore F_{gmdy} > F_{gmsy}$~~

~~$\therefore a_y < 0$ for $8M$~~

~~$F_{net\,y} = F_{gmdy} - F_{gmsy}$~~

2) $M_s = 4M + 2M$,

$M_d = 8M$, $M_2 = 4M+2M+8M$

$F_{gmsy} = g \cdot \sin\theta \cdot M_s$

$F_{gmdy} = F_{Md} = g \cdot 1 \cdot M_d$

$\because 1 \geq \sin\theta$

& $M_d > M_s$

$\therefore F_{Md} > F_{gmsy}$

$\therefore$ acceleration for $8M$ is downward

$F_{net} = F_{Md} - F_{gmsy}$

$F_{net} = gM_d - g\sin\theta M_s$

$= g\,8M - g\sin\theta\, 6M$

$= g\,2M(4 - \sin\theta \cdot 3)$

$F_{net} = M_2 a$

$a = \frac{F_{net}}{M_2}$

$= \frac{g\,2M(4-\sin\theta\cdot 3)}{14M}$

$= \frac{g(4-3\sin\theta)}{7}$ (positive because $g = -9.81\ m/s^2$)

b) $T_2 = 4M_2$

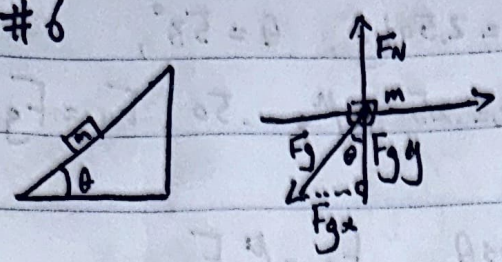
$T_1 = 8M_2$

$T_1 - T_2 = 8M_2 - 4M_2$

$= 4M_2$

$= 4M \frac{g(4-3\sin\theta)}{7}$

#6



Given: $m = 2.50\,kg$, $\theta = 53°$, $\mu_k = 0.25$, $\mu_s = 0.50$

$F_N = -F_{gy}$
$= -mg\cos\theta$

$f_s = \mu_s F_N$
$= -\mu_s mg\cos\theta$

$f_k = \mu_k F_N$
$= -\mu_k mg\sin\theta$

a) $F_{gx} = mg\sin\theta$
$= 2.5\,kg \cdot 9.81\,m/s^2 \cdot \sin 53°$
$= -19.586\,N$

$F_s = 7.380\,N$
$F_k = 3.690\,N$
$F_s <$ magnitude of F_{gx}

$F_{net} = 0$

$F_{net} = F_{gx} + f_k$
$0 = mg\sin\theta + \mu_k F_N$
$-mg\sin\theta = \mu_k m a$
$a = -\frac{g\sin\theta}{\mu_k m}$
$= +12.535\,m/s^2$

b) $F_{net} = F_{gx} + f_k$
$\cancel{ma = \frac{mg\sin\theta + \mu_k m a_k}{m}}$
$\cancel{= g\sin\theta + \mu_k a_k}$
$\cancel{=}$

$-ma = -19.586\,N + -3.690\,N$
$-a = \frac{-23.276\,N}{2.5\,kg}$
$= -9.3104\,m/s^2$

c) $F_{net} = F_{gx} + f_k$
$-ma = -19.586\,N + 3.690\,N$
$-a = -\frac{15.896\,N}{2.5\,kg}$
$= 6.3584\,m/s^2$

#7

Given: $m_A = 2.00\,kg$, $m_B = 3.00\,kg$, $\mu_s = 0.250$

a) $\because$ m_B is moving at ~~cons~~ constant velocity,

$\therefore a = 0$

$f_s = \mu_s F_N$

$\because a = 0$

$F = ma$

$\therefore F = 0$

$\therefore f_s = 0$

b) $f_{s,max} = \mu_s |F_N|$

$= \mu_s |m_A g|$

$= .25 \cdot |2.00\,kg \cdot -9.81\,m/s^2|$

$= 4.90\,N$

$m_A a = 4.90\,N$

$a = \frac{4.90\,N}{2.00\,kg}$

$= 2.45\,m/s^2$

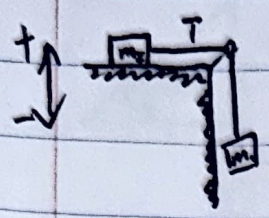
$F = m_{all}\, a$

$= (3.00\,kg + 2.00\,kg) \cdot a$

$= 5\,kg \cdot 2.45\,m/s^2$

$= 12.25\,N$

#8



Given: $m_1 = .820\,kg$, $m_2 = 1.40\,kg$, $\mu_k = .200$

$T = m_1 g$
$= -.820\,kg \cdot 9.81\,m/s^2$
$= \cancel{-9} -8.044\,N$

$F_k = \mu_k F_N$
$= .200 \cdot 13.734\,N$
$= 2.7468\,N$

$F_N = -m_2 g$
$= -1.40\,kg \cdot -9.81\,m/s^2$
$= 13.734\,N$

$m_a = m_1 + m_2$
$= .82\,kg + 1.4\,kg$
$= 2.22\,kg$

$F_{net} = m_1 g + f_k$
$= -8.044\,N + 2.7469\,N$

$a = -\frac{5.2972\,N}{m_a}$
$= -\frac{5.2972\,N}{2.22\,kg}$
$= -2.386\,m/s^2$